

Name: _____

Hammack

Score: 12/12**Directions:** Put away phones, calculators, smart watches, etc.

Answer in the space provided.

Show work.

1. Use any method discussed in class to find $\begin{vmatrix} 2 & -1 & -2 & 5 \\ 3 & 3 & 6 & 1 \\ 0 & 1 & 3 & 0 \\ -2 & 4 & 8 & -2 \end{vmatrix}$.

To introduce some zeros, first do $C_3 - 2C_2 \rightarrow C_3$. Then expand along the third column:

$$\begin{vmatrix} 2 & -1 & -2 & 5 \\ 3 & 3 & 6 & 1 \\ 0 & 1 & 3 & 0 \\ -2 & 4 & 8 & -2 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 0 & 5 \\ 3 & 3 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -2 & 4 & 0 & -2 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 5 \\ 3 & 3 & 1 \\ -2 & 4 & -2 \end{vmatrix}$$

To get a quick zero, do $R_1 + R_3 \rightarrow R_3$. Then expand along the first column.

$$\begin{vmatrix} 2 & -1 & 5 \\ 3 & 3 & 1 \\ -2 & 4 & -2 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 5 \\ 3 & 3 & 1 \\ 0 & 3 & 3 \end{vmatrix} = 2 \begin{vmatrix} 3 & 1 \\ 3 & 3 \end{vmatrix} - 3 \begin{vmatrix} -1 & 5 \\ 3 & 3 \end{vmatrix} = 2 \cdot 6 + 3 \cdot 18 = \boxed{66}.$$